# KATHERINE DAVEY

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## PROFESSIONAL EXPERIENCE (9 YEARS)

### **Lead QA Automation Engineer** | New York Life Insurance **05/2025 – Present**

#### Leadership

Owned end-to-end testing for enterprise-wide initiatives spanning Corporate Communications, Employee Engagement, Finance, CoE.

- Led functional testing and UAT across 3 departments, coordinating with external software vendors to clarify requirements, establish a requirements schema, and implement a standardized defect tracking system.
- Identified and mitigated a critical SSO authentication defect in production within 24 hours of go-live, preventing sustained impact to external customers.
- Presented proof-of-concepts and AI roadmap recommendations to VP-level department leadership.
- Gathered and translated technical and business requirements from stakeholders into actionable AI agent testing strategies.
- Improved process documentation and test planning standards across the QA team.
- Created automated, visual API workflows using Postman's Flow's feature to enable teams to test data pipelines across core platforms and understand user data flow across platforms.
- Designed a comprehensive, reusable strategy to test AI integrations across complex, enterprise platforms.
- Collaborate with product owners, vendor evaluation and selection, coordinating across product owners, developers, third-party service teams, and UAT testers to ensure cohesive strategy.

#### AI Agent Evaluation & Testing

Led functional and integration testing of third-party agentic AI solutions.

- Collaborated with platform and IAM engineers, solution architects, and vendors to build and test agentic AI workflows on no/low-code platforms, resulting in successful integrations of third-party test strategy across vendors.
- Developed AI evaluation strategy, defining key metrics and standardized pass/fail criteria to validate third-party AI chatbot agents for enterprise HR deployment, ensuring teams had accurate user data across complex, enterprise platforms.
- Created custom GPTs using prompt engineering and explicit input/output criteria to support evaluation consistency.
- Led vendor evaluation and selection, coordinating 30+ team members of product owners, developers, third-party service teams, and UAT to ensure cohesive strategy.

#### Automation & Tooling

Spearheaded shift-left testing and automation, aligning development and business on quality goals earlier in the delivery cycle.

- Built a test automation framework & Python tooling to automate data validation, reducing timeline from 1 week to minutes.
- Developed a C#/NUnit data validation tool that ingests Excel files, performs column header validation, and outputs a structured discrepancy report packaged and distributed via JFrog Artifactory for team-wide use.
- Wrote shell scripts to automate file-based content migration between learning platforms, reducing manual effort for a large-scale content transfer initiative.
- Used Playwright to automate bulk file downloads, replacing a time-intensive manual process.
- Automated end-to-end browser workflows using Mabl (AI test automation platform) to reduce repetitive manual testing.
- Spearheaded shift-left testing initiative, aligning development and business teams on quality goals earlier in the delivery cycle.

### **Software Development Engineer in Test (SDET) |** Anthology Inc. **11/2021 – 05/2025**

Led test automation for Blackboard Learn Ultra's Gradebook module, a large-scale React SPA serving millions of students globally.

- Engineered tooling to enable reliable UI testing of React components using Selenium-Java, solving dynamic SPA challenges.
- Enhanced test framework to support Data-Driven Testing (DDT) by integrating RestAssured API calls into end-to-end test cases.
- Implemented BDD with Cucumber to bridge communication between engineering, business, and management stakeholders.
- Leveraged AI (Aqua IDE) to debug failing legacy tests and auto-generate documentation, accelerating a large codebase modernization effort.
- Configured Jenkins and GitHub Actions CI/CD pipelines to automate builds and streamline release cycles.
- Performed frontend performance analysis with Google Lighthouse and backend load testing with JMeter.
- Mentored junior engineers in automation scripting, framework patterns, and code review best practices.
- Presented automation demos and QA reports to product owners and clients, maintaining strong stakeholder relationships.
- Performed complex SQL database testing to verify data accuracy and integrity, resolving 95% of data discrepancies pre-deployment.

## OTHER PROJECTS

### LLM Evaluator

Used DeepEval / OpenAI / Python to build a backend service to programmatically compare LLM responses against baseline results. Users submit prompts, expected outputs, and select evaluation metrics; the OpenAI client manages the request-response loop and DeepEval hosts structured results. Defined evaluation categories and metrics using DeepEval's framework to standardize AI output quality scoring. Built a Streamlit frontend with Cursor to surface evaluation results accessibility for non-technical stakeholders.

### Environmental Monitoring & Control System (IoT)

Designed and built a closed-loop environmental monitoring and control system using Arduino (C++) and sensors. Integrated soil moisture, temperature, humidity, and light sensors with automated actuator control (fan, water pump, LED) via a WiFi-connected web dashboard supporting real-time bidirectional communication reading sensor state and sending control commands from the browser to the device. Implemented WiFi networking to enable remote real-time monitoring and device control without physical connection.

**ETL Engineer & Data Scientist** | CUNY Advanced Science Research Center

### Embedded Device Data Logging

Developing a database and repository to organize 10+ years of clinical research data, including an ingestion pipeline for health and environmental data collected from Fitbit devices and third-party APIs into a SQL database to support active research.Created user workflow and data mapping diagrams to illustrate and clarify data storage locations, connections, and access permissions. Collaborate with interdisciplinary stakeholders with varying degrees of technical knowledge to gather requirements, define project scope, identify available resources/restrictions, and understand current data pipelines and procedures. Completed specialized training to ensure compliance with Human Subject Research (HSR) and HIPAA regulations. Developing schemas, cleaning and reorganizing legacy data to ensure consistency and usability for ongoing research and analysis using Sharepoint, SQL, Python.

### Automated Calling System

Research assistants dreading weekend on-call rotations asked me to develop an automated calling and response logging system. Used Twilio ML Studio and Zapier to create automated workflow integrating Google Calendar, Mail APIs, and custom webhooks. Successfully implemented a service to trigger an automated call 24 hours before a scheduled appointment, prompt the participant to 'confirm' or 'cancel' via digital input, log the response on Google Calendar and send an automated email to team members.

## TECHNICAL SKILLS

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| **Programming Languages** | Java, C#, JavaScript, SQL, Bash/Shell, Python, HTML/CSS, C ++ |
| **Testing Frameworks & Tools** | Playwright, Selenium, Appium, Postman, BrowserStack, SauceLabs, RestAssured, Cucumber, TestNG, JUnit, NUnit, Jest, ReqnRoll, X-Ray, Zephyr, JMeter, Chrome DevTools, Google Lighthouse |
| **Automation Platforms & Tools** | Power Automate, Power Query, Zapier, Twilio, SAP Build |
| **AI Tools** | DeepEval, OpenAI, Mabl AI, Claude Code, GitHub Copilot, Cursor, ChatGPT |
| **CI/CD & Infrastructure** | GitHub Actions, Jenkins, TeamCity, Git, Github, Gitlab, Bitbucket |
| **Other** | Docker, AWS (ECR/ECS/EC2/ELB/CloudWatch), Jira, Confluence, JFrog Artifactory, SharePoint, Agile Scrum, Kanban, Shift-Left Testing, Test Driven Design, Behavior Driven Design (BDD), Data Driven Testing |

## SELECTED PUBLICATIONS


Ramjan S, Blum M, Tseng RY, **Davey K**, Shereen AD, Nomura Y. Feasibility of Integrating Wearable Devices and Ecological Momentary Assessment for Real-Time Environmental Exposure Estimation: Proof-of-Concept Study. JMIR Form Res. 2026 May 8;10:e86615. doi: 10.2196/86615. PMID: 42102276; PMCID: PMC13155499.

Nomura, Y., Ramjan, S., Wang A. , Blum M., Brabander C., DeIngeniis D., **Davey K.**, Bailey R., Hinton V., Sheffield P. Prenatal Heat and Air Pollution Exposure Alter Progesterone and Child Behavior: A Longitudinal Study.

Nomura, Y.; Kittler, P.; Taveras, S.; Sie, S.Y.; Nelson, E.; **Davey, K.**; Kaushal, R.; Rajendran, K.; Gordon, A.; Shereen, D.; et al.. Psychological Effects of COVID-19 on Pregnant Women and New Mothers Living in a US Hotspot. https://europepmc.org/article/ppr/ppr368725.

Nomura Y, **Davey K**, Pehme P M, Finik J, Glover V, Zhang W, Huang Y, Buthmann J, Dana K, Yoshida S, Tsuchiya K J, Li X, Ham J.. Influence of in-utero exposure to maternal depression and natural disaster-related stress on infant temperament at 6 months: The children of Superstorm Sandy. Infant Mental Health Journal, 40(2), 204-216. https://doi.org/10.1002/imhj.21766.

Buthmann, J., Ham, J., **Davey, K.**, Finik, J., Dana, K., Pehme, P., Zhang, W., Glover, V., & Nomura, Y.. Infant Temperament: Repercussions of Superstorm Sandy-Related Maternal Stress.Child psychiatry and human development, 50(1), 150–162. https://doi.org/10.1007/s10578-018-0828-2.

Zhang W, Rajendran K, Ham J, Finik J, Buthmann J, **Davey K**, ... & Nomura Y.. Prenatal exposure to disaster-related traumatic stress and developmental trajectories of temperament in early childhood: Superstorm Sandy pregnancy study. Journal of Affective Disorders, 234, 335-345. https://doi.org/10.1016/j.jad.2018.02.067.


## EDUCATION

**B.A. Classical Languages (Linguistics)** — Hunter College, City University of New York

Minor in Digital Communication